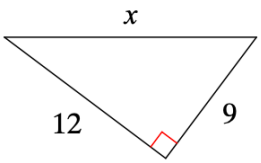
GEOMETRY

Chapter 9 Test Form A

SHOW ALL WORK TO RECEIVE FULL CREDIT!

Solve the following triangles for the missing side. Answer in reduced radical form.

1.



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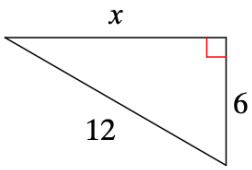


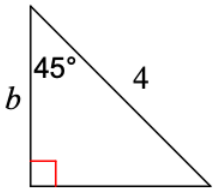
Identify the triangles as right, acute, or obtuse. Justify your answer given the following lengths of the sides.

3. 12, 14, 15

4. 19, 9, 12

Use your special right triangles for problems #5-9. Answer in <u>reduced radical form</u> and do not use a calculator.

5.



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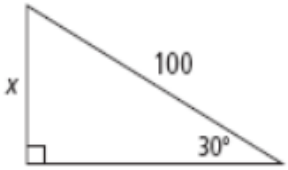

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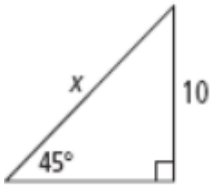

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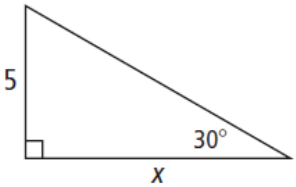

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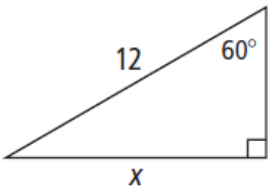

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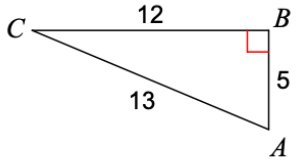
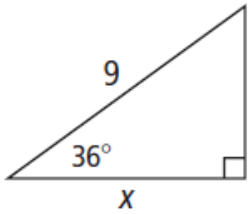
10. Write each ratio from the triangle as a reduced fraction.

a. sin A **b. cos A** **c. tan A** **d. tan C**



For problems 11 – 15, find the value of x or the missing angle to the nearest tenth.

11.



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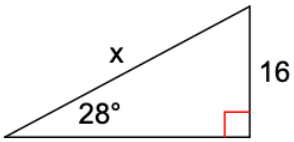

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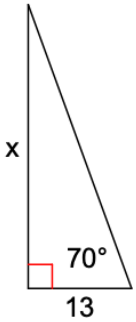

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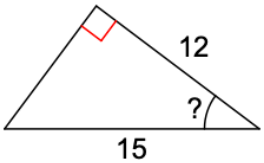

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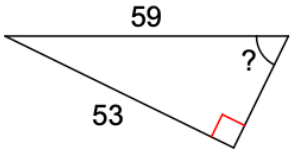
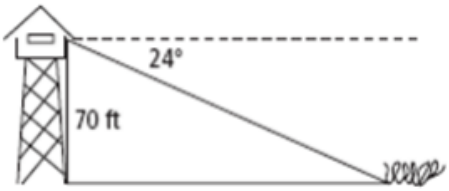


16. A fire ranger stands at an observation window 70 feet above the ground. She sees a fire in the distance. She takes a reading of the angle of depression and finds it to be 24 degrees. To the nearest tenth of a foot, how far away is the fire from the base of the tower?



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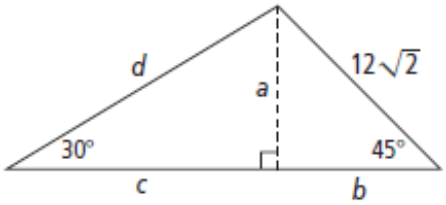
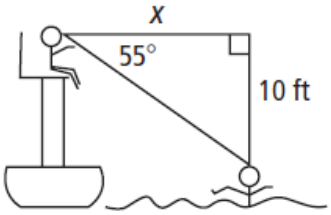
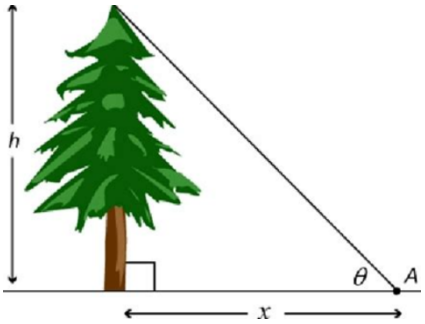
17. Find the measures of *a, b, c*, and *d* in the figure and give your answers in reduced radical form.



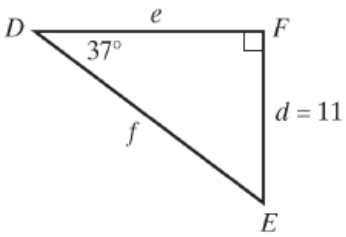
18. Suppose you are a lifeguard looking down at a swimmer in a swimming pool at a 55° angle of depression. You are 10 ft up in your seat. How far is the swimmer from the base of the lifeguard stand? Round your answer to the nearest tenth.



19. Given the height of a tree is 135 feet. Find the angle of elevation to the top of the tree from a point that is 40 feet from the base of the tree to the nearest tenth.



20. Solve the following triangle. Round any decimal answers to the nearest tenth.



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